AI 101 for Mechanical Engineers · Module 15

Verification and validation are different checks

Lesson 15.2

What you will learn

> Separate implementation verification from real-system validation, choose suitable evidence for each, and keep enough provenance for another engineer to reproduce and audit the procedure.

Lesson overview

Module 15 · Lesson 2

Verification and validation are different checks

AI 101 for Mechanical Engineers

1 Verification

Did we implement the method correctly?

- Known cases and limiting cases
- Dimensional analysis and bounds
- Sign conventions
- A workflow that can be repeated

$$P_h = \Delta p \times Q \rightarrow Pa \times m^3/s = W$$

Hydraulic power (Δp = pressure rise, Q = flow rate): a unit check is verification. It catches coding errors but does not show the model represents a real pump.

2 Validation

Does it represent the real system for its purpose?

- Independent, calibrated measurements
- Across the relevant operating conditions
- Evidence kept apart from fitting and model selection

Matching synthetic data or a simulator checks the code and workflow only. It does not validate real hardware.

Not the same as the "validation set" of Module 10, which is used for model selection.

3 Provenance and reproducibility

Record so another engineer can repeat it:

- How and when data were acquired
- Sensor and calibration
- Units and transformations
- Split identifiers
- Code and environment versions
- Model settings and manual decisions

Reproducible is not correct: the same bug gives the same wrong answer every time. A seed helps repeat randomized steps; it does not validate the model.

4 Re-check after a change

New sensor · Recalibration · New operating regime · New plant → review calibration, defer unsupported cases, and collect new validation data or retrain by stated criteria, with named reviewers.

Key takeaway

Verification checks the implementation; validation checks the real system.

Neither a passing synthetic test nor a reproducible run replaces physical evidence.

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Verification and validation ask different questions

Verification

- Was the method implemented correctly? [13]
- Known cases, dimensional analysis, bounds, limiting cases, signs

Validation

- Is the model adequate for its purpose? [13]
- Independent, calibrated measurements from the real system
- Across relevant operating conditions

This differs from the Module 10 validation set, which supports model selection.

A unit check verifies the pump power product

$$\Delta p \times Q$$
$$\mathrm{Pa} \times \mathrm{m}^3/\mathrm{s} = \mathrm{W}$$

- Pressure rise times volumetric flow rate
- Unit and known-calculation tests can catch implementation errors
- They do not show the model represents a real pump

Synthetic agreement is not physical validation

- Synthetic data or simulators can test code and workflow
- Agreement with synthetic data alone gives no physical validation
- Keep final validation evidence independent of fitting and selection
- Preserve the final test boundary until development is complete

Record provenance for every result

Record	Examples
Data acquisition	How and when; sensor and calibration context
Processing	Units, transformations, split identifiers
Software and model	Code and environment versions, settings, manual decisions

> Record these [36], [38] so others can regenerate the procedure and trace supporting evidence.

Reproducible does not mean correct

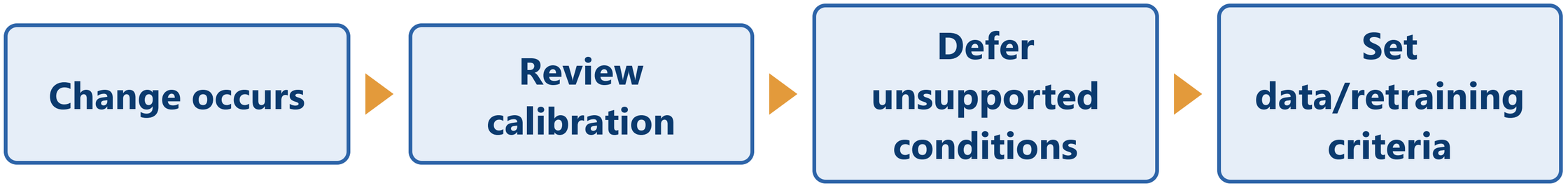
Reproducibility

- Another person can regenerate the procedure
- The same bug can repeat the same wrong answer

Random seed

- Can make randomized steps more repeatable
- Does not guarantee identical results across environments
- Does not validate a model

Changes to the system trigger a re-check



- Triggers: new sensor, recalibration, operating regime or plant
- Set explicit criteria for new validation data or retraining
- Keep named reviewers and auditable records

CHECK YOUR UNDERSTANDING

Is matching a synthetic simulator enough to validate predictions for real equipment?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Is matching a synthetic simulator enough to validate predictions for real equipment?

> No. It can verify the code and evaluation workflow. Validation needs independent, calibrated real-system measurements across the relevant operating conditions.

CHECK YOUR UNDERSTANDING

A unit check confirms that Δp × Q gives watts. Is that verification or validation?

Answer before moving on.

CHECK YOUR UNDERSTANDING

A unit check confirms that Δp × Q gives watts. Is that verification or validation?

> Verification. It checks the implementation, not whether the model represents a real pump.

Key takeaways

- Verification: implemented correctly; validation: represents the real system
- Synthetic or simulator agreement does not validate real equipment
- Keep final validation evidence independent of fitting and selection
- Record provenance for reproduction and audit; reproducible is not correct
- Re-check after new sensor, recalibration, regime or plant

References

[13] W. L. Oberkampf and C. J. Roy, Verification and Validation in Scientific Computing. Cambridge, U.K.: Cambridge Univ. Press, 2010. doi: 10.1017/CBO9780511760396

[36] J. Pineau et al., “Improving reproducibility in machine learning research (A report from the NeurIPS 2019 reproducibility program),” J. Mach. Learn. Res., vol. 22, no. 164, pp. 1–20, 2021.

[38] T. Gebru et al., “Datasheets for datasets,” Commun. ACM, vol. 64, no. 12, pp. 86–92, 2021. doi: 10.1145/3458723

Numbers match the course reference list. The written lesson is the authoritative text.